

**Input Clinical Data
CDC BRFSS 2015**

- N = 253,680 Samples
- 21 Clinical Features
 - Target: Diabetes
- Sensitive: Income/Edu

Stage 1: Adaptive Clustering Engine

- Multi-Objective Criterion:
 $\text{Composite}(K) = \text{DB}(K) + 0.5 \cdot \text{Demog. SD}(K)$
- Candidate Algorithms: KMeans, Auto, MiniBatch, GMM
 - Candidate Cluster Count: $K \in [2, 10]$
- Isolates Distinct Demographic & Clinical Sub-Populations
(e.g., Low-Risk Affluent vs High-Risk Impoverished)

Stage 2: Heterogeneous Classifier Engine

- Stratified Cross-Validation per Sub-Population
 - Dynamic Model Assignment:
 - LightGBM / XGBoost (Non-linear complex)
 - Random Forest (Ensemble)
 - Logistic Regression (Parametric linear)
- Autonomous Match to Local Data Geometry

Global search objective: $\min f(n) = (1 - \text{AUC}) + \lambda \cdot \text{DPD} + h(n)$